

# Sex, gender and equity in Medicare reimbursement

## Preliminary evidence synthesis and research agenda

Coverage, service valuation and access to care for women and men

September 16, 2026

### The central question

Do Medicare benefit rules, payment values and practical barriers align with the clinical needs of women and men? This paper identifies signals worth testing, including possible disadvantages affecting either sex, while distinguishing patient access from physician payments.

### Abstract

The selected evidence suggests that reimbursement inequity is a question about clinical indication, benefit design, service valuation and completed care. It does not establish a uniform disadvantage to one sex. The formulary snapshot and basic Part D exclusion rules contradict the broad premise that Medicare excludes estradiol while routinely covering Viagra for erectile dysfunction. Procedure studies report differing results, and shared conditions such as osteoporosis reveal possible gaps affecting men. Physician payment studies address a separate question.

This paper brings together 34 retained literature records and 40 selected care-item/indication rows, describes two reproducible CMS data summaries, and proposes eight exploratory hypotheses. The Lancet review by Mauvais-Jarvis and colleagues provides the organizing clinical framework. [L-B01; L-B01-E1]

**Review status:** Preliminary narrative synthesis. Comprehensive searching, final eligibility adjudication, full-text extraction, formal risk-of-bias appraisal and outcome-specific certainty assessment remain unfinished. No meta-analysis or causal effect estimate is presented.

Prepared from the Medicare research repository: [github.com/clinical-trials/medicare](https://github.com/clinical-trials/medicare). No institutional affiliation or external endorsement is asserted.

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# What the evidence currently supports

The working thesis is that inequity can arise when policy and practical access fail to meet clinical need. A difference in spending or use is a signal for investigation; its cause and clinical significance require separate evidence.

## 1. Formulation and indication matter

Generic oral estradiol and vaginal cream had a listed option without prior authorization or step therapy in all 5,499 analyzed plan segments. Other formulations varied. Basic Part D excludes drugs when used for sexual or erectile dysfunction; other eligible indications and supplemental benefits must be assessed separately. [P-002; P-003; P-026]

## 2. Procedure comparisons are sensitive to the match

One study reports lower female-anatomy work RVUs in 41 of 55 pairs, while a smaller, closely matched study finds no statistically significant overall difference. Our five-pair 2026 illustration contains differences in both directions. [L-P01; L-P02; P-035; P-036]

## 3. Sex-specific care can be assessed on its own merits

A benefit change can be evaluated against desired, clinically indicated care within one population. An opposite-sex counterpart is not required. Observational contraceptive coverage evidence provides one starting point. [L-G09]

## 4. Shared-condition gaps may disadvantage men

Historical bone-density testing was much lower in men. A selected male fracture cohort benefited from outreach, although the trial did not compare men with women. Current eligibility includes several qualifying routes. [L-G04; L-G05; P-012]

## 5. Physician payments are a distinct evidence stream

Selected studies report lower annual Medicare payments to women physicians. Service volume, referrals, clinical time, specialty and billing circumstances require investigation; some per-service comparisons are null. [L-C01; L-C02; L-C03]

## 6. Coverage alone is an incomplete access measure

Financial, transport and caregiving barriers may affect treatment completion. Interviews and clinical frameworks suggest mechanisms, but do not quantify Medicare-specific causal effects. [L-Q01; L-B01]

# Methods: design and eligibility

## Design, objective and review status

This preliminary narrative synthesis combines a selected literature register, Medicare policy review and descriptive analyses of public CMS files. The question is whether coverage rules, service values, patient costs and administrative requirements align with clinically indicated care for women and men. The project has not completed a systematic review. No prospective registration or protocol fixed before searching is documented; the eight hypotheses were developed after evidence inspection.

## Medicare benchmark and population

Medicare serves as a national policy benchmark for later-life care. Adults aged 65 and older accounted for 17% of the population and 37% of all-payer personal health care spending in 2020. These figures do not establish that Medicare finances most US care. Younger beneficiaries remain eligible, including those entitled through disability, end-stage renal disease or amyotrophic lateral sclerosis; age and entitlement strata should be preserved when available. [P-041; P-043]

At the investigator's direction, the population scope was amended on September 16, 2026, after initial evidence identification, to focus on cisgender women and men. Transgender-focused patient studies and gender-affirming-care indications were excluded from this project. Male/female reporting alone does not establish gender identity; uncertain cohorts remain candidates pending adjudication. Mixed-population findings require separately extractable eligible results. Policy and code evidence is assessed by indication without inferring patient identity.

## Source identification and access

The register was assembled through investigator-provided publications and targeted source retrieval. Article material came from PubMed, PMC, publisher or author-institution pages, indexed text and selected PDFs. Policy sources included federal statutes, CMS manuals and coverage documents, Medicare.gov, CMS datasets and identified contractor or plan examples. The Lancet review guided clinical framing and citation following; its cited references were not automatically treated as screened evidence. [L-B01; L-B01-E1]

September 16, 2026 is the extraction and verification date, not the endpoint of a completed reproducible database search. Complete search strings, search-specific result counts, universal deduplication and a full screening flow are not documented. No comprehensive language, publication-date or database coverage can therefore be claimed. National rules, local decisions and individual plan examples retain their distinct scope.

# Methods: evidence handling

## Selection, extraction and classification

The active register retains empirical patient studies and code/policy components as candidate evidence; clinical reviews, qualitative accounts and physician-payment studies have separately labeled roles. Exclusion reasons and original broad-search records remain in the audit trail. Independent duplicate human screening, duplicate extraction and final adjudication have not been completed.

Structured records capture citation, design, population, payer, data period, findings, limitations, relevance and source access where recorded. Titles, authors, DOI and PMID were checked against PubMed/MEDLINE metadata, including Europe PMC MEDLINE records. PMCID and PMID remain distinct; corrections are linked rather than counted as new studies. Extraction combines abstracts, accessible excerpts and targeted main/full-text review. Metadata verification is not confirmation of every finding or final eligibility.

Each care row is classified by shared systems/functions, female or male reproductive anatomy, female or male reproductive physiology, or mixed/site-dependent services. Each study separately records patient, service, physician or general-background subject; analytical unit; comparison axis; and inference limits. The framework requires clinical need to be defined independently of coverage rules; patient-level need was not measured in these CMS summaries. Figure 1 presents these distinctions.

| Register component     | Count and interpretation                                                                                                       |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Care items             | 40 selected care-item/indication rows; not a census or representative sample.                                                  |
| Active literature      | 34 records: 27 primary candidates or code-policy components; 7 contextual records.                                             |
| Analytic subjects      | 18 patient care/access; 10 service/anatomy valuation; 3 physician sex/gender; 3 general clinical background.                   |
| Separate audit records | 8 excluded studies, 3 excluded care items and 1 linked correction. Two methodology references are outside the 34-record count. |

## Narrative synthesis and appraisal

Findings were grouped by indication, unit, payer, policy period and outcome. Coverage, RVUs, actual payments, patient spending and completed care were kept separate. Official rules describe policy, not effects. Null, reversed and conflicting findings were retained. No meta-analysis, causal estimate or national prevalence of bias was calculated. Limitations were recorded descriptively; formal design-specific risk-of-bias, publication-bias and outcome-specific GRADE assessments remain unfinished. [M-002]

AI-assisted retrieval, extraction and scripted checks supported this work; computational verification does not replace independent expert screening or appraisal. Register counts are not a PRISMA flow diagram. Planned systematic-review procedures are described later and are not represented as completed methods. [M-001]

# Methods: CMS data analysis

## Formulary cohort and drug matching

Figure 2 uses the August 2026 CMS monthly file (release August 26; archive member label August 31) and the August 25 formulary reference file, retrieved September 16. Repeated county rows were collapsed to unique contract/plan/segment identifiers; a consistency check required one formulary per segment. Rows whose suppression flag was not N were excluded: 19 distinct segments. The 112,294 source rows yielded 5,499 unsuppressed segments, 5,055 contract-plan pairs and 328 formularies. CMS public-file scope omits employer, PACE, most demonstration and non-Part-D plans. [P-002; P-034; P-037; P-038]

Basic formulary entries were joined to the reference by RxNorm concept identifier (RxCUI) and to segments by formulary ID. Target groups used ingredient, dosage form and brand fields. Single-ingredient hormone groups excluded combination concepts; generic formulation groups used non-brand-specific clinical-drug concepts (TTY = SCD). Estring was identified by brand. Listing required at least one matching product/strength, not every dose, manufacturer or package. Twenty-three unmatched basic-file RxCUIs remained unclassified; no eligible formulary was missing.

## Three mutually exclusive states per formulation

PA = prior authorization; ST = step therapy. The same option must have neither flag to meet state U.

| State                   | Operational definition                                                                         |
|-------------------------|------------------------------------------------------------------------------------------------|
| U: option without PA/ST | At least one matched row has both prior-authorization and step-therapy flags equal to N.       |
| R: all options flagged  | One or more matched rows are listed, but none has both flags equal to N: listed count minus U. |
| NL: no matched listing  | No matched row is listed: 5,499 minus listed count.                                            |

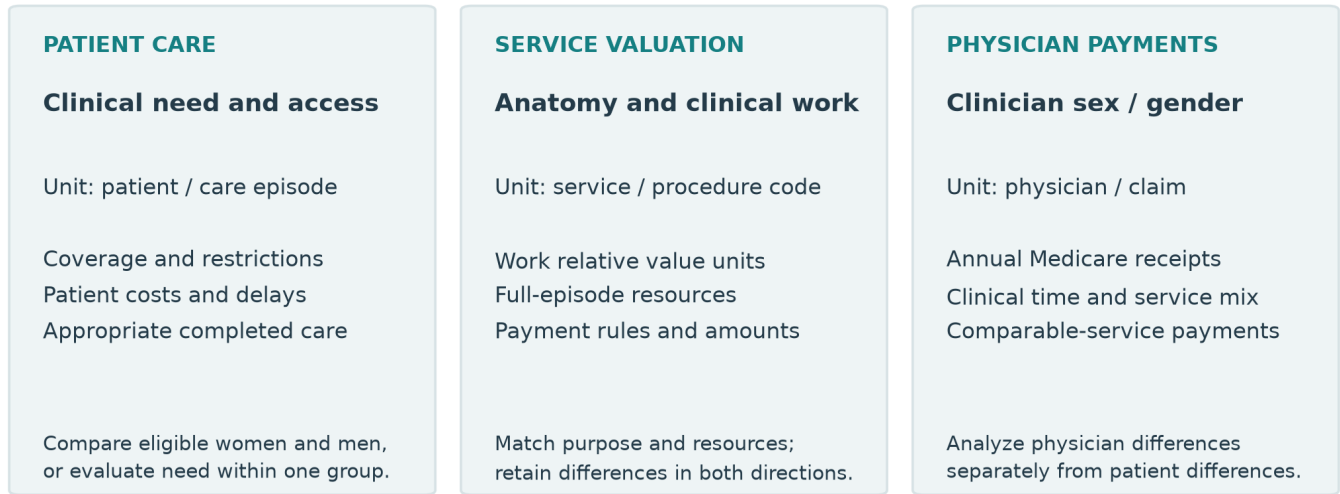
Each segment receives equal weight. Counts sum to 5,499 per row; percentages equal count / 5,499 x 100, rounded for display. Ten specific groups are plotted; two overlapping all-form aggregates are omitted. No enrollment weighting, confidence intervals or between-drug significance tests were applied. Reused formularies are not independent policy decisions. Flags do not establish clinical suitability, actual criteria, costs, denials or the outcome of an exception.

## Procedure values and computational checks

Figure 3 uses five purposively selected published procedure pairs and the July 2026 CMS RVU26C nonQPP file, released June 30. We extracted active-status, blank-modifier code rows and computed first-code minus second-code work RVUs. Values are unadjusted; no geographic adjustment, dollar conversion or pooled effect was calculated. Published matching is not proof of equal complete work; code 51597 is not male-exclusive. [L-P02; P-035; P-036; P-045]

Scripts checked joins, denominators, state sums and all five RVU differences. A separate AI-assisted implementation reproduced selected formulary joins/counts. Source files, extraction scripts, concept lists and validation records are retained. Figures use Matplotlib; the PDF uses ReportLab. No patient-level claims or identifiable records were analyzed.

# Separate the clinical and payment questions



**Anatomy, patient sex/gender and physician sex/gender are separate variables.**

**Figure 1. Three analytical streams.** Proposed classification framework, not a measured causal model. The patient, service and physician units must remain separate. Clinical framing draws on the Lancet review and its correction. [L-B01; L-B01-E1]

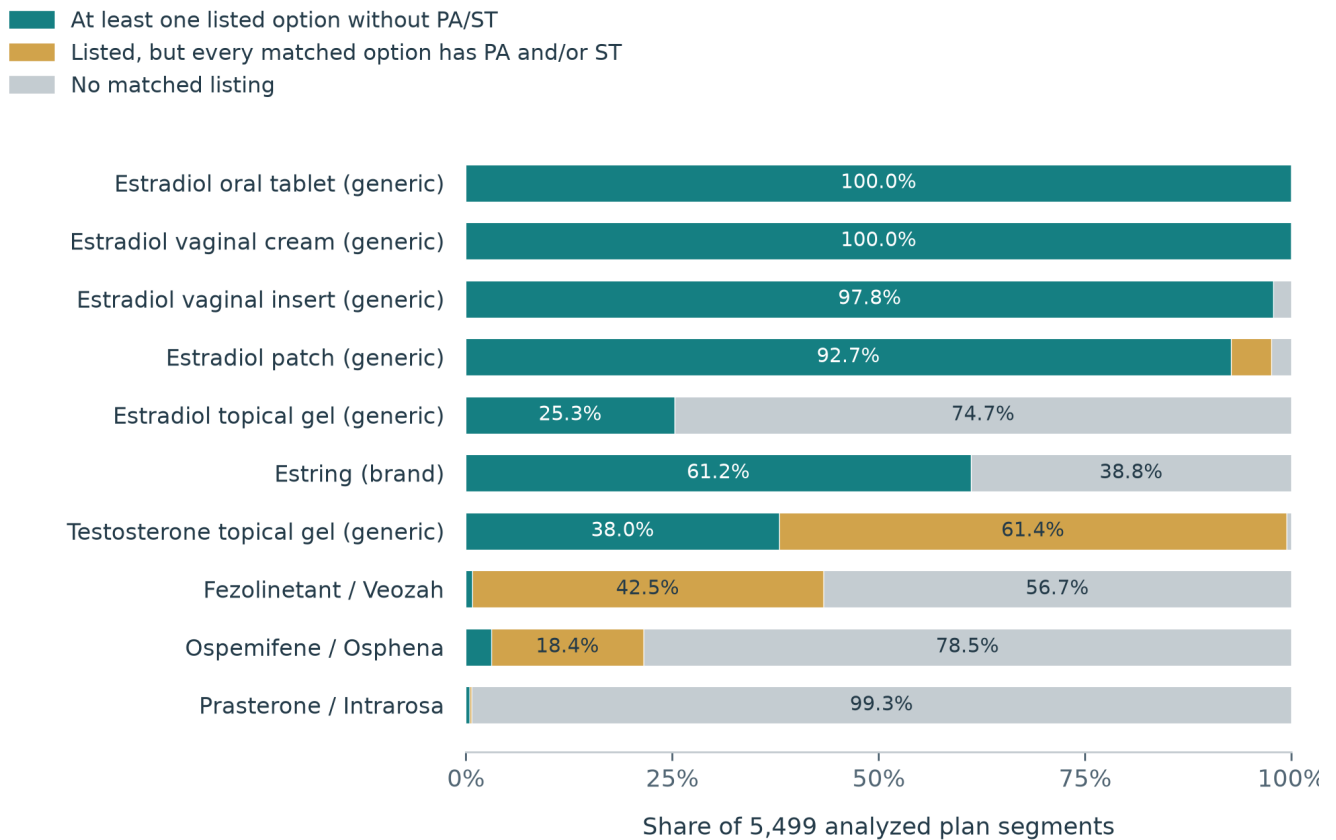
Anatomy and physiology describe the care being delivered. They do not establish a person's gender identity or the gender of the clinician. Record the original study's sex/gender terminology and measurement method.

| Clinical basis                | Examples and comparison approach                                                                                                                                                               |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shared body systems/functions | Bone, breast, lymphatic, cardiovascular and pelvic-floor care. Compare need within the same indication; sex-defined eligibility or unequal disease burden does not make the anatomy exclusive. |
| Female reproductive anatomy   | Local vaginal or cervical care. Assess clinical need, benefits, alternatives and the policy rationale.                                                                                         |
| Male reproductive anatomy     | Prostate or penile care. Use the same need-based assessment; do not assume an interchangeable female counterpart.                                                                              |
| Reproductive physiology       | Menopause and male hypogonadism are distinct indications. Hormone names alone do not define clinical equivalence.                                                                              |
| Mixed/site-dependent services | Split bundled procedures or services by site, indication and code before comparison.                                                                                                           |

Clinical need must be defined independently of coverage eligibility. Otherwise people excluded by a rule disappear from the denominator, concealing possible unmet need.

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# Drug coverage varies within clinical care



**Figure 2. Formulary states for ten selected formulations, August 2026.** Each bar contains 5,499 unsuppressed plan segments, equally weighted, using 328 formularies. PA = prior authorization; ST = step therapy. States are mutually exclusive within each row. Small segments are unlabeled; exact counts appear in the appendix. Sources: CMS monthly file and reference file. [P-002; P-034; P-037; P-038]

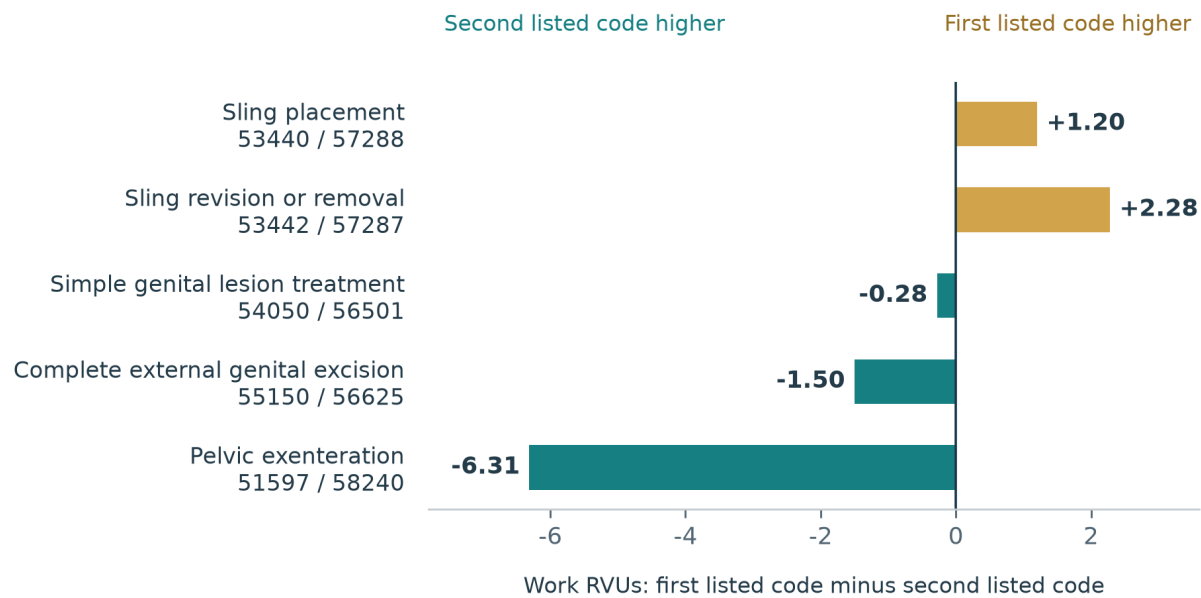
The national snapshot does not support a blanket estradiol exclusion. Oral, vaginal and transdermal products still differ in indication, dose, suitability and restrictions. Testosterone gel is shown as another clinical context, not an equivalent treatment comparator.

Basic Part D excludes a drug when used for sexual or erectile dysfunction. A product used for another eligible indication, such as sildenafil for pulmonary arterial hypertension, requires a separate coverage assessment. Brand-versus-generic naming does not establish inequity. [P-003; P-026]

Clinical evidence is mixed: one vaginal estradiol trial found no primary symptom advantage over dual placebo; a broader review found possible benefits for selected symptoms, generally with low certainty. These findings do not settle every formulation or indication. [L-D03; L-D04]

**Inference limit:** Listed means at least one matched product/strength. These data contain no beneficiary enrollment weights, clinical need, actual costs, denials or treatment outcomes. PA/ST flags do not disclose the criteria. Nonlisting does not rule out a successful exception. [P-033]

# Procedure valuation: signals in both directions



**Figure 3. Five selected procedure pairs, July 2026.** Bars show national, unadjusted work relative value units (RVUs): first listed code minus second. Published anatomical/indication comparisons are illustrative, not random or a pooled sex effect. The urologic exenteration code 51597 is not exclusive to male patients. [P-035; P-036; P-045]

| Study                  | Observed result                                                                         | What it can establish                                                                             |
|------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Penn, 2025 [L-P01]     | 41/55 female-anatomy codes had lower work RVUs; male codes averaged 30% higher.         | A selected-pair valuation signal; incomplete full-text extraction.                                |
| Hathaway, 2024 [L-P02] | 7/10 female procedures had higher work RVUs; no significant overall difference.         | Sensitivity to clinical matching; a small null result is not equivalence.                         |
| Polan, 2021 [L-P03]    | Female procedures: 10.6 vs 9.7 RVUs/hour, but \$555 vs \$599 modeled compensation/hour. | Different measures can give different directions; dollars use specialty compensation assumptions. |

Work RVUs are not dollars paid. Complete comparisons need independently measured preoperative, intraoperative and postoperative work, intensity and risk. Matching global periods alone is insufficient. For the genital-lesion pair, compare like treatment methods; the male code specifies chemical treatment. [P-035; P-036]

Breast reconstruction adds a within-condition question: flap procedures may have lower work RVUs per operative hour than implant procedures. This is a complexity/valuation question, not a patient-sex or physician-gender estimate. [L-P07]

# From benefit rules to completed care



**Include people with unmet need and those who never generate a paid claim.**

**Figure 4. Proposed care-pathway measurement framework.** Arrows organize measurements; they do not represent estimated causal effects. Clinical need is assessed independently of reimbursement eligibility. Practical barriers are hypotheses to test. [L-B01; L-Q01]

## Shared-condition care can reveal gaps affecting men

In historical Medicare data from 1999-2005, 30.0% of women and 4.4% of men in the analyzed white/Black populations received central dual-energy X-ray absorptiometry (DXA) at least once. These are utilization findings, not need-adjusted estimates of discrimination. In a later trial of selected men after fragility fracture, outreach increased testing from 4.9% to 10.7% and treatment initiation from 2.5% to 4.0%. That trial had no female comparison. [L-G04; L-G05]

## A policy change can be studied within one population

Among younger women with disabilities, gaining dual Medicare-Medicaid coverage was associated with a 3.9-percentage-point increase in contraceptive use (95% CI, 3.5-4.3). This observational result motivates study of desired, appropriate contraception; more utilization is not automatically better care. [L-G09]

## Historical burden must be interpreted against current policy

A breast-cancer study reported greater monthly out-of-pocket spending with lymphedema. Insurer-policy research also describes coverage gaps. Both predate Medicare's compression-treatment benefit, which began in 2024. A current evaluation should test unmet need and costs after the change; compression items, pumps and surgery are distinct benefits. [L-A01; L-A02; P-021; P-022]

Kidney-care interviews with 51 nephrologists in 22 countries suggest financial, transport and caregiving barriers. Only seven respondents were in the United States. Their accounts are perceptions, not measured Medicare denial rates or causal estimates. [L-Q01]

# Physician sex/gender: a separate question

The user-supplied ophthalmology, cardiology and surgery papers compare physicians. They cannot establish that women patients receive lower coverage or payments. Annual Medicare receipts also differ from physician salary, per-service payment and patient out-of-pocket cost. [L-C01; L-C02; L-C03]

| Evidence                          | Outcome and finding                                                                           | Interpretation boundary                                                                                  |
|-----------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Ophthalmology Ahmad, 2020 [L-C01] | Women ophthalmologists had lower adjusted annual Medicare collections.                        | Abstract-level extraction; collections do not measure salary or a patient-sex contrast.                  |
| Cardiology Raber, 2021 [L-C02]    | Women cardiologists had lower annual payments; common-code comparisons included null results. | Service volume and mix matter. Annual differences do not establish different rates for identical claims. |
| Surgery Munir, 2024 [L-C03]       | Adjusted annual payments were lower for women in general, oncologic and colorectal surgery.   | CPT matching cannot recover all modifiers, settings or clinical circumstances.                           |

## A concrete illustration from surgery

For general surgeons, the adjusted annual female-minus-male payment difference was -\$14,963 (95% CI, -\$18,822 to -\$11,105). In colorectal surgery, the unadjusted payment-per-service comparison was not statistically significant (P = .52). These are different outcomes and analyses; they should not be pooled into a single claim about reimbursement rates. [L-C03]

## A testable physician pathway

Referral opportunities, procedure opportunities and clinical time may contribute to annual receipts. Measure the total annual difference first, then examine these pathways. Automatically adjusting for every pathway may remove part of the difference under investigation. Comparable-rate testing requires claim-level information on service, modifier, place of service, geography and billing circumstances.

### Keep the question attached to its unit

Patient access: did an eligible person receive appropriate care?

Service valuation: how is comparable clinical work valued?

Physician payments: how do receipts differ across clinicians and why?

The physician stream is contextual to the primary patient-access inquiry. It remains separate in figures, evidence extraction and future certainty assessments.

# Exploratory hypotheses: patient access

These hypotheses were developed after inspecting the selected evidence. They are research propositions, not preregistered analyses or confirmed findings. Clinical need and patient preferences must be established independently of coverage rules.

## HYP-01 / Formulation restrictions

**Hypothesis:** For beneficiaries who need a particular formulation for menopause symptoms or genitourinary syndrome of menopause (GSM), greater cost or authorization burden reduces timely use of suitable treatment.

**Test:** Link within-indication plan changes to orders, denials, initiation, persistence and symptoms; assess clinical suitability and enrollment weighting. **What would weaken it:** Precise evidence of no meaningful access difference after accounting for safety, alternatives and preferences. [L-D01; L-D03; L-D04; P-002; P-029; P-033]

## HYP-03 / Contraceptive benefit design

**Hypothesis:** More comprehensive benefits improve receipt of desired, appropriate contraception among younger beneficiaries with disabilities.

**Test:** Study verified coverage transitions with comparison groups, clinical need and stated preferences; distinguish drug and procedure benefits. **What would weaken it:** A precise lack of meaningful improvement despite a verified reduction in financial or administrative barriers. [L-G09; L-G10; P-015; P-018; P-019]

## HYP-04 / Bone care after fracture

**Hypothesis:** Men receive less appropriate osteoporosis care after fracture than women with comparable need, and targeted outreach reduces this gap.

**Test:** Use a contemporary eligible cohort, including nonrecipients; measure referrals, testing, treatment, follow-up and competing mortality. **What would weaken it:** Precise need-adjusted parity, or precise failure of an effective outreach intervention to reduce the proposed gap. [L-G04; L-G05; P-012]

## HYP-05 / Practical barriers in kidney care

**Hypothesis:** Financial, transport and caregiving burdens contribute to a sex difference in completing indicated kidney care.

**Test:** Measure burdens and completion directly, then evaluate support that actually reduces the between-sex burden difference. **What would weaken it:** Such support precisely fails to reduce the completion gap. Absence of a sex-by-intervention interaction alone is not a refutation. [L-Q01]

# Exploratory hypotheses: valuation and policy

## HYP-02 / Comparable clinical work

**Hypothesis:** Across a prespecified matched set, female-anatomy procedures have lower average work RVUs after accounting for independently measured complete clinical work.

**Test:** Fix the clinical matches before viewing values; measure the full episode. Estimate mean paired differences and matching sensitivity. **What would weaken it:** Precise average parity would weaken the average-penalty hypothesis, without proving every individual pair fair. [L-P01; L-P02; L-P03; P-035; P-036]

## HYP-06 / Physician opportunity and receipts

**Hypothesis:** Referral and procedure opportunities and clinical time contribute to annual Medicare receipt differences by physician sex/gender.

**Test:** Estimate the total annual difference first; examine time-standardized and pathway-specific estimates second, with complete claim circumstances. **What would weaken it:** Precisely estimated minimal contribution from the proposed pathways, or disappearance of the annual difference under comparable observation. [L-C01; L-C02; L-C03]

## HYP-07 / The 2024 compression benefit

**Hypothesis:** The Medicare compression-treatment benefit reduces cost-related unmet need among beneficiaries with lymphedema.

**Test:** Use post-policy patient data, a defensible comparator and pretrend checks; measure costs and receipt of needed compression items. **What would weaken it:** An estimate precise enough to exclude a meaningful reduction in unmet need; an imprecise null is inconclusive. [L-A01; L-A02; P-021; P-022]

## HYP-08 / Complex reconstruction

**Hypothesis:** Complex flap reconstruction has lower valuation relative to complete clinical work than implant reconstruction, beyond similar patterns in other reconstructive services.

**Test:** Measure complete episodes and outside comparators first. Study patient access separately when recipient and nonrecipient data are available. **What would weaken it:** A sufficiently precise estimate excluding a prespecified meaningful relative disadvantage. Broad real-payment declines alone are insufficient. [L-P07; L-P08; L-P09; P-035]

Priority sequence: complete the formulation audit; extend the analogous-procedure set; design a contemporary shared-condition patient-access study. All eight hypotheses remain untested in this project.

# Completing the systematic review

## Search and screen transparently

The next phase should specify dated searches across bibliographic databases and Medicare policy sources, publish complete search strings, deduplicate records, and use independent screening with documented resolution of disagreements. Expand beyond reproductive care to cardiovascular, kidney, lung, cancer, metabolic, musculoskeletal, pain, mental-health and preventive care. Use the Lancet review for citation following, then screen underlying papers independently. [L-B01]

## Appraise evidence at the appropriate level

Assess risk of bias with tools matched to study design. Extract policy authority, effective date, population, indication, payer, comparator, effect estimate, uncertainty and source access. Separate national rules from contractor and plan implementation; separate Original Medicare, Medicare Advantage and Part D. Prespecify causal questions and avoid automatically adjusting away possible mechanisms of inequity.

| Method         | Plain-language purpose                                                                                                                             | Current status                                                           |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| PRISMA [M-001] | Show how a review identified, selected, assessed and synthesized evidence. It is reporting guidance, not a quality score.                          | A complete reproducible search and screening flow are not yet available. |
| GRADE [M-002]  | Assess confidence in the body of evidence for a specific outcome, considering bias, inconsistency, indirectness, imprecision and publication bias. | No formal outcome-specific certainty ratings have been completed.        |

## Limits of this white paper

The evidence set is selected and concentrated in reproductive care. Eligibility and extraction remain incomplete for some studies; several reports have only abstract or excerpt access. Historical findings span different policy eras. Formulary data are not enrollment weighted, and reused formularies make plan segments dependent. Procedure pairs are purposive and do not measure all clinical work. Claims alone miss untreated need, cash purchases and some denials. These constraints prevent a national estimate of the prevalence or causal magnitude of reimbursement bias.

### Plain-language concept for public communication

We are asking whether Medicare rules help women and men obtain the care they actually need. We will track how evidence is found and selected using PRISMA, then assess how much confidence each important conclusion deserves using GRADE. The goal is to identify where policies appear fair, where care may be harder to obtain, and what evidence is still missing. This paper is the starting synthesis, not the completed review.

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# Data appendix: figures 2 and 3

**Table A1.** Exact plan-segment counts underlying Figure 2. Every row totals 5,499. U = at least one listed option without PA/ST; R = listed, but every matched option has PA and/or ST; NL = no matched listing. These states are not rates of claim approval. [P-002; P-034; P-037; P-038]

| Formulation                        | U    | R    | NL   |
|------------------------------------|------|------|------|
| Estradiol oral tablet (generic)    | 5499 | 0    | 0    |
| Estradiol vaginal cream (generic)  | 5499 | 0    | 0    |
| Estradiol vaginal insert (generic) | 5380 | 0    | 119  |
| Estradiol patch (generic)          | 5096 | 270  | 133  |
| Estradiol topical gel (generic)    | 1391 | 0    | 4108 |
| Estring (brand)                    | 3364 | 0    | 2135 |
| Testosterone topical gel (generic) | 2089 | 3376 | 34   |
| Fezolinetant / Veozah              | 42   | 2338 | 3119 |
| Ospemifene / Osphena               | 171  | 1012 | 4316 |
| Prasterone / Intrarosa             | 24   | 15   | 5460 |

**Data handling.** County repetitions were removed from 112,294 plan-county rows. Nineteen suppressed segments were excluded. The resulting 5,499 segments comprise 5,055 contract-plan pairs and reuse 328 formularies. CMS public-file exclusions include employer, PACE, most demonstration and non-Part-D plans. Formulation groups were matched through the August reference file; 23 basic-file RxCUIs were unmatched and unclassified. PA/ST criteria, quantity limits, dose suitability, patient costs and enrollment weights are not evaluated here.

**Table A2.** Work RVUs underlying Figure 3. National unadjusted values, July 2026 nonQPP file; no geographic or claim-specific adjustments. Columns follow the published comparator order, without inferring patient sex. [P-035; P-036; P-045]

| Code pair (first / second) | First code | Second code | Difference |
|----------------------------|------------|-------------|------------|
| 53440 / 57288              | 13.03      | 11.83       | +1.20      |
| 53442 / 57287              | 13.15      | 10.87       | +2.28      |
| 54050 / 56501              | 1.26       | 1.54        | -0.28      |
| 55150 / 56625              | 7.94       | 9.44        | -1.50      |
| 51597 / 58240              | 41.79      | 48.10       | -6.31      |

Reproducibility: figures derive from research/synthesis\_analysis.json and research/procedure\_pairs\_2026.json. The repository preserves source dates, URLs, data extraction scripts and source notices. A static release is dated September 16, 2026; later policy or formulary changes require a new analysis.

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## References and source notes

Stable IDs connect this paper to the evidence workbook and living bibliography. Every cited scientific article has a verified PMID below. Access notes describe the extraction available to this project, not the quality of the publication. Policy labels are descriptive source labels. No excluded study supplies this synthesis.

**[L-A01]** Dean LT, Ransome Y, Frasso-Jaramillo L, Moss SL, Zhang Y, Ashing K, et al. Drivers of cost differences between US breast cancer survivors with or without lymphedema. *J Cancer Surviv.* 2019;13(5):804-814. PMID: [31446591](#). DOI: [10.1007/s11764-019-00799-1](#).

Project access: Citation/abstract extraction recorded; full-text review pending.

**[L-A02]** Lynn JV, Hespe GE, Akhter MF, David CM, Kung TA, Myers PL. Cross-Sectional Analysis of Insurance Coverage for Lymphedema Treatments in the United States. *JAMA Surg.* 2023;158(9):920-926. PMID: [37285151](#). DOI: [10.1001/jamasurg.2023.2017](#).

Project access: Citation/abstract extraction recorded; full-text review pending.

**[L-B01]** Mauvais-Jarvis F, Bairey Merz N, Barnes PJ, Brinton RD, Carrero JJ, DeMeo DL, et al. Sex and gender: modifiers of health, disease, and medicine. *Lancet.* 2020;396(10250):565-582. PMID: [32828189](#). DOI: [10.1016/s0140-6736\(20\)31561-0](#).

Project access: Targeted main-text review; citation and linked correction verified.

**[L-B01-E1]** Department of Error. *Lancet.* 2020;396(10252):668. PMID: [32891210](#). DOI: [10.1016/s0140-6736\(20\)31827-4](#).

Project access: Correction metadata verified.

**[L-C01]** Ahmad S, Ramulu P, Akpek E, Deobhakta A, Klawe J. Gender-Specific Trends in Ophthalmologist Medicare Collections. *Am J Ophthalmol.* 2020;214:32-39. PMID: [31926887](#). DOI: [10.1016/j.ajo.2019.12.024](#).

Project access: Abstract verified; full text not reviewed.

**[L-C02]** Raber I, Al Rifai M, McCarthy CP, Vaduganathan M, Michos ED, Wood MJ, et al. Gender Differences in Medicare Payments Among Cardiologists. *JAMA Cardiol.* 2021;6(12):1432-1439. PMID: [34495296](#). DOI: [10.1001/jamacardio.2021.3385](#).

Project access: Targeted main-text review via indexed PMC; supplement not reviewed.

**[L-C03]** Munir MM, Dillhoff M, Tsai S, Collins C, Dedhia P, Pawlik TM. Gender-Based Variations in Medicare Reimbursements Among Different Surgical Subspecialties. *JAMA Surg.* 2024;159(9):1060-1070. PMID: [39046733](#). DOI: [10.1001/jamasurg.2024.2298](#).

Project access: Targeted main-text and tables review; supplement not independently reviewed.

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